Spain-Argentina 2026 World Cup Final

A one-second Polymarket top-of-book event study

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Abstract

This paper studies a locally recorded, one-second top-of-book data set covering 98 Polymarket outcome tokens during the 2026 World Cup final. We combine public minute-level match anchors with synchronized movement in a compact set of core markets to infer likely information-arrival seconds. The analysis finds distinct market reaction bursts around kickoff, halftime, the late decisive sequence, the winner, and full time. Definitive events generated larger and more persistent repricing than ambiguous events, while probability coherence among the three regulation outcomes remained tight. Results are descriptive: public commentary clocks and exchange timestamps are not synchronized, and midpoint changes are not a claim of executable arbitrage or causal price discovery.

Keywords: prediction markets, market microstructure, event studies, football, order books, information arrival

Data coverage: 2026-07-19T17:59:37.094+00:00 to 2026-07-19T22:04:10.890+00:00; 1,437,660 compact top-of-book snapshots. The core analysis uses 14 token series across seven linked match markets.

1. Introduction

Prediction-market prices are attractive as high-frequency summaries of publicly available information, but their interpretation around sporting events is difficult. A market can react to the on-pitch event, a broadcast signal, an official decision, a commentary update, or a change in quoted liquidity. The central problem in this study is therefore modest: given a local one-second record and only approximate event minutes, identify the market seconds that most plausibly correspond to new information and describe the associated order-book behavior.

Research questions

The paper asks five questions. (1) Which core markets move together around major match events? (2) Are ambiguous events more reversible than definitive ones? (3) Does quoted top-of-book liquidity withdraw or expand around large moves? (4) Do regulation-outcome prices temporarily violate their probability-sum relationship? (5) Does a small lead-lag exercise reveal a stable one-second information leader? The objective is interpretability, not a universal trading model.

Contribution

The contribution is a reproducible local workflow: raw WebSocket data are retained as compact snapshots, event anchors are explicitly treated as windows, and every derived feature is computed in pandas after SQL filtering. This design is intentionally simpler than a large vector autoregression over the full market universe, which would be difficult to interpret in a single match.

2. Polymarket and data collection

Polymarket markets are binary token pairs. The collector subscribed without authentication to the public CLOB market WebSocket at wss://ws-subscriptions-clob.polymarket.com/ws/market using asset IDs, sent the required PING heartbeat every 10 seconds, maintained a local book from book snapshots and price-change updates, and wrote one top-of-book observation per token per second. The research data set stores best bid, best ask, their sizes, midpoint, spread, initialization state, and whether the top of book changed. It deliberately does not store every raw message or continuous full depth.

Collection design

Item	Design choice	Why it matters
Recorder coverage	98 tokens	The paper restricts attention to a transparent 14-token core sample.
Sampling	One compact snapshot per token per second	Balances temporal resolution with manageable storage.
Book state	Local book from snapshots plus deltas	Avoids treating sparse best-price updates as a complete book.
Persistence	SQLite WAL, batched inserts	Provides durable local research data without per-row commits.

Core analytic sample

The paper selects seven high-information markets: Spain regulation win, Argentina regulation win, draw in regulation, team to advance, match extra time, over/under 0.5, and both teams to score. Both YES and NO tokens are retained, yielding 14 series and 205,380 rows. This is small enough for transparent event windows while preserving logically linked outcomes.

3. Event timeline and identification strategy

FotMob provides useful minute-level ticker anchors, but it does not supply a clock synchronized to the exchange record. We cross-checked the important sequence against FIFA, AP, El Pais, and Al Jazeera. The timeline establishes broad UTC search windows rather than hard timestamps. The extended halftime entertainment is a practical reason to widen post-break windows.

Inferred-information rule

For each event window, we calculate one-second midpoint changes for the 14 core series. Each absolute change is divided by a robust pre-window scale (the 90th percentile of absolute changes in the preceding three minutes), capped to prevent one stale quote from dominating, and summed across series. The highest-scoring second is the inferred arrival. This is a transparent detector of a synchronized quote response, not an oracle for event time.

Event	Minute	Inferred UTC	Score	Series moving
Kickoff	0'	19:00:11	21.0	4
Messi early chance	6'	19:12:00	24.5	8
Halftime	45'	19:44:29	31.0	8
Second-half kickoff	46'	20:25:07	21.0	4
Enzo Fernández red card	90+2'	21:09:56	57.2	12
Williams goal disallowed	96'	21:09:56	57.7	12
Torres winning goal	106'	21:26:24	20.0	2
Full time	120+'	21:42:28	20.0	2

A key result is non-separation: the red-card and disallowed-goal search windows select the same 21:09:56 UTC movement burst. The correct conclusion is not that one event caused it, but that the available price record lacks enough independent timing information to allocate the burst cleanly.

4. Main analysis: separating a late disturbance from final resolution

The most informative contrast in this record is between the late 21:09:56 burst and the later 21:26:24 winner burst. In the former, the team-to-advance binary pair moves 14.3 cents and retains almost all of that move after one minute. At the same second, Spain regulation and draw prices each make roughly 7.7-7.9 cent excursions, but more than 90% of those excursions reverse inside the next minute. In other words, the market distinguishes a durable change to the eventual-advancement path from a provisional or mechanically refreshed repricing of the regulation-only contracts.

This is precisely the kind of difference a headline chart can hide. The shared red-card/disallowed-goal candidate is not merely a large movement: it is a cross-market sorting event. Advancement prices retain the information, while regulation and draw quotes largely mean-revert as the game moves beyond their economically relevant horizon. Displayed top size simultaneously expands by about 35% for the advancement pair but by more than 200% in Spain regulation, consistent with book replenishment after a short-lived regulation shock rather than a persistent probability reset.

For the 106-minute winner, the detector identifies a distinct later response. By then, many regulation, scoring, and total markets no longer have usable midpoint observations; only the team-to-advance pair provides a clean continuously quoted outcome. Its immediate peak move is about 16.1 cents. The selected arrival score is lower because only two core series remain active, not because the football event is economically weak. This is an important lifecycle effect in the data: a cross-market event detector must be interpreted alongside the number of live quotes.

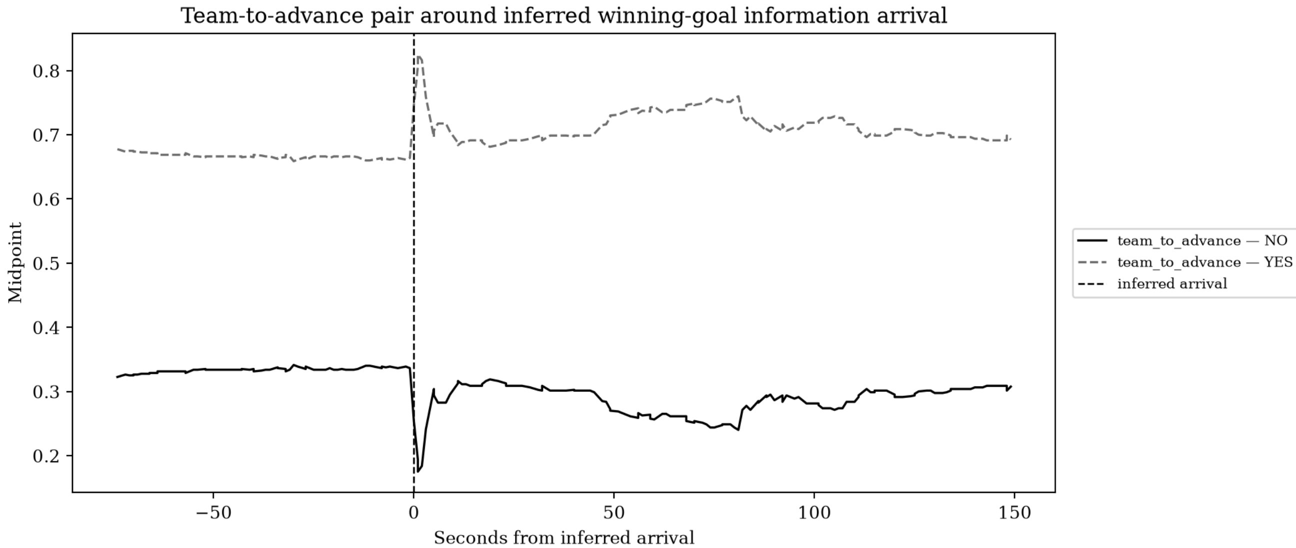

Figure 1. Team-to-advance pair around the inferred winning-goal information arrival. The vertical line is selected by synchronized movement, not the public commentary timestamp.

Interpretation

The eventual full-time transition reinforces the difference between information and settlement. The advancement pair then moves roughly 32.3 cents with less than 1% reversal, while its displayed top size falls about 73%. That is the expected signature of resolution: price converges and the incentive to maintain two-sided liquidity disappears. The paper therefore reports peak response, end-of-minute persistence, and liquidity change together instead of elevating a single midpoint observation into a complete causal account.

5. Ambiguity, reversal, and liquidity

For each inferred arrival, the baseline is the mean midpoint in the previous 30 seconds. We measure the largest absolute deviation in the following minute, the remaining deviation at one minute, and define reversal as one minus the absolute remaining deviation divided by the peak deviation. The same before/after intervals produce a simple percent change in aggregate best-bid plus best-ask size.

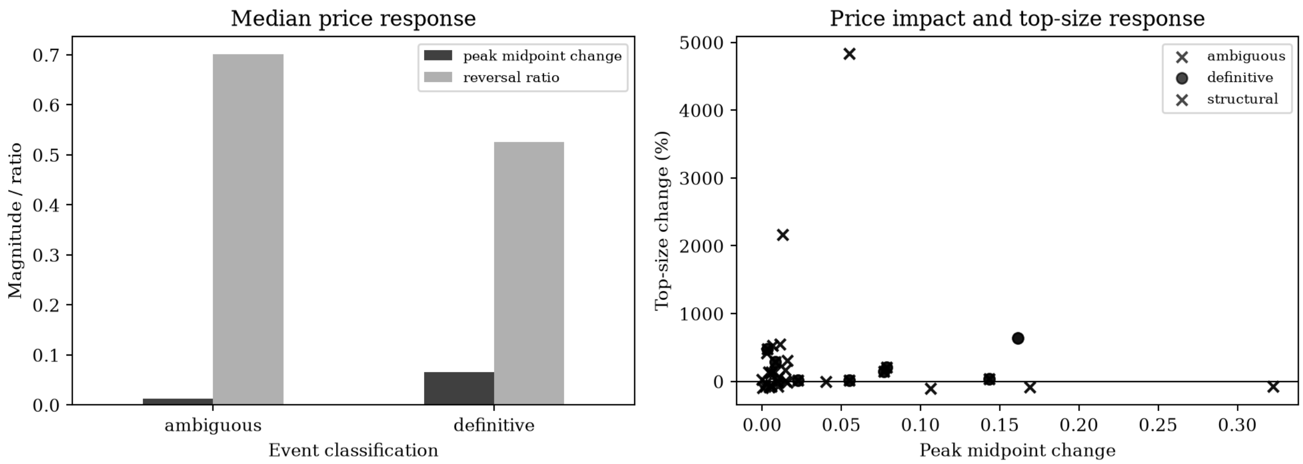

Figure 2. Median response statistics and individual liquidity observations. The small number of events calls for descriptive reading rather than formal inference.

Event class	Median peak move	Median reversal	Median top-size change
Ambiguous	0.012	70.1%	84.8%
Definitive	0.066	52.6%	12.8%

Ambiguous events have a smaller median initial response (roughly 1.2 cents) but a higher median reversal ratio (roughly 70%) than definitive events (roughly 6.6 cents and 53%). Liquidity does not follow a universal withdrawal story: some uncertainty windows show more top size after the burst, while the winner is approximately flat in aggregate. This highlights a central microstructure caveat: midpoint changes must be read jointly with displayed size and spread.

6. Logical-market consistency

The three regulation YES outcomes - Spain win, draw, and Argentina win - form a simple probability triangle. Their midpoints should approximately sum to one. We use the residual as a quote-synchronization diagnostic, not as a conclusion about risk-free arbitrage. An executable strategy would require depth, fees, order priority, fill risk, and latency that the compact snapshot data do not establish.

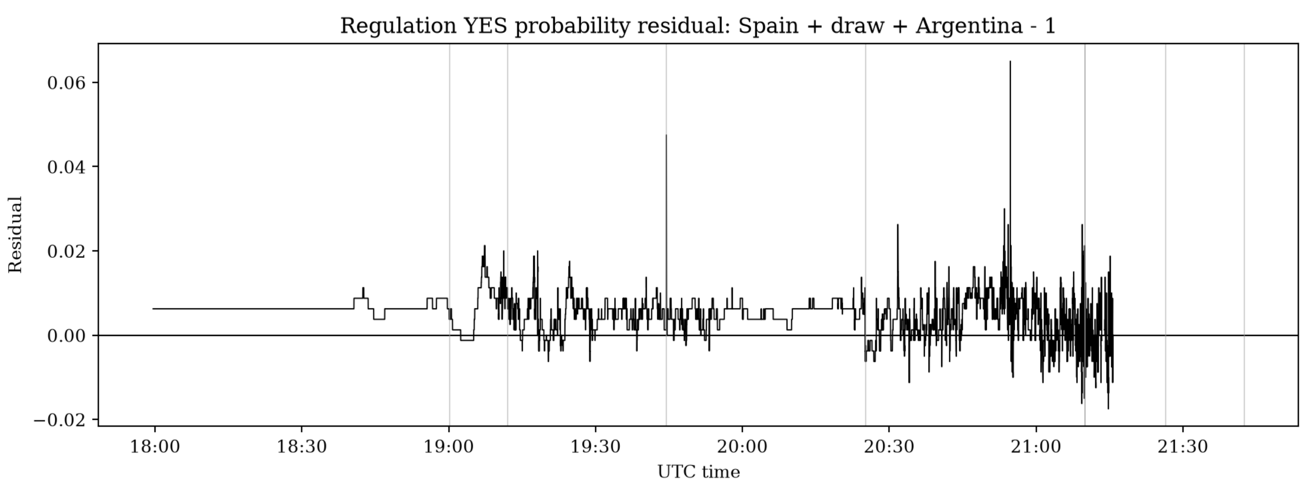

Figure 3. Regulation probability-sum residual. Thin vertical marks are inferred event-arrival seconds.

Quantile	Absolute Spain + draw + Argentina residual
50th percentile	0.0063
90th percentile	0.0088
99th percentile	0.0162

The residual is typically small: the median absolute error is about 0.63 cents and the 99th percentile is about 1.62 cents. Short-lived departures are plausible under asynchronous quoting and differing displayed sizes. The result is most useful as a quality-control and event-diagnostics plot.

7. Lead-lag and a compact regression

A high-dimensional VAR across all markets would overstate what one match can identify. Instead, we examine only Spain regulation YES and team-to-advance YES around the late burst while both markets are active. Positive lag in Figure 4 means the regulation series moved first. The regression uses one-second target changes explained by contemporaneous and one-second-lagged source changes plus the target's own lag.

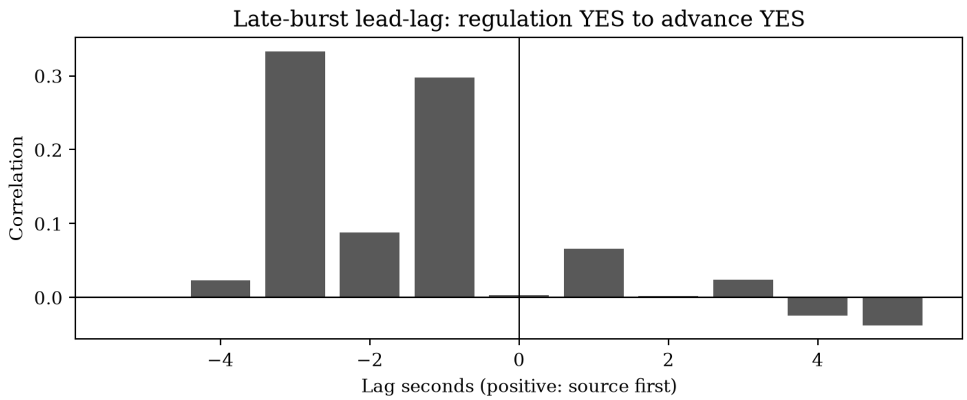

Figure 4. Descriptive lag correlations over a 90-second window around the late burst.

Term	Coefficient	Std. error	t statistic
intercept	0.0001	0.0001	1.12
source_change_t	-0.0079	0.0268	-0.29
source_change_t_minus_1	0.0217	0.0263	0.83
target_change_t_minus_1	0.0815	0.0341	2.39

The evidence is not a stable causal leader. The strongest association is contemporaneous or near-contemporaneous, and the target's own lag has the most prominent coefficient. That pattern is consistent with synchronized reaction and mechanical quote updating. It should not be read as evidence that one market can be traded ahead of the other.

8. Experiments and robustness ideas

The following extensions are deliberately actionable rather than speculative:

Experiment	Implementation	What it would clarify
Clock sensitivity	Shift each public event window by plus/minus 30 seconds and re-run detection.	Whether an inferred arrival is robust to public-clock uncertainty.
Depth-aware response	Record several book levels and quote age in a future collector.	Whether apparent midpoint moves had executable depth.
Cross-source timing	Add synchronized broadcast/video or official event-feed timestamps.	Separate the red card, review, and disallowed-goal information sequence.
Spread conditioning	Stratify event moves by pre-event spread and top size.	Whether liquid books react differently from thin books.
Out-of-sample matches	Repeat the exact pipeline on other knockout matches.	Whether ambiguity and reversal patterns generalize.

Practical lessons for future collection

The current collector is well matched to exploratory event studies: it preserves every second's top of book while avoiding raw-message storage. Future improvements should prioritize synchronized external event timestamps, a small amount of depth beyond the best level, and explicit receipt-time diagnostics before adding complex statistical models.

9. Limitations

The sample is one match, public commentary minutes are delayed and approximate, and the data are top-of-book snapshots rather than trades or full executable depth. The market may react to television latency, official review, crowd knowledge, or another public feed. Pairwise regression coefficients are therefore descriptive. The report does not offer investment advice, a trading rule, or a causal claim about information flow.

10. Reproducibility and appendix

The notebook provides the narrative analysis, while the small modules in 01-experiment separate data access, event timing, movement detection, metrics, plotting, and the paper build. The appendix below lists the only seven markets used in this paper, together with their original selection notes.

Markets used in this paper

Analytic label	Full market question	Selection note
argentina_regulation	Will Argentina win on 2026-07-19?	Tracks regulation result, advancement, extra time, and penalties.
both_teams_score	Spain vs. Argentina: Both Teams to Score	Threshold ladder reveals cross-market consistency after goals.
draw_regulation	Will Spain vs. Argentina end in a draw?	Tracks regulation result, advancement, extra time, and penalties.
extra_time	Spain vs. Argentina: Will the Match Go to Extra Time?	Tracks regulation result, advancement, extra time, and penalties.
over_0_5	Spain vs. Argentina: O/U 0.5	Threshold ladder reveals cross-market consistency after goals.
spain_regulation	Will Spain win on 2026-07-19?	Tracks regulation result, advancement, extra time, and penalties.
team_to_advance	Spain vs. Argentina: Team to Advance	Tracks regulation result, advancement, extra time, and penalties.

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